

Zhiyi Shi

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Education

University of Illinois Urbana-Champaign

Champaign, IL, US

PhD in Computer Science

09/2025 – 05/2028 (expected)

- Advisor: Prof. Jimeng Sun & Prof. Quanzheng Li (Harvard MGH)
- Research Interests: Multimodal Learning, VLM, Parameter Efficient Fine-Tuning

Carnegie Mellon University

Pittsburgh, PA, US

MS in Computer Science

09/2022 – 05/2024

Southeast University

Nanjing, China

BS in Computer Science

09/2018 – 06/2022

Selected Papers | [Full List](#) |

- TPAMI 2026** Chongjie Si*, **Zhiyi Shi***, Shifan Zhang, Xiaokang Yang, Hanspeter Pfister, Wei Shen.
Task-Specific Directions: Definition, Exploration, and Utilization in Parameter Efficient Fine-Tuning.
- COLM 2025** **Zhiyi Shi**, Binjie Wang, Chongjie Si, Yichen Wu, Junsik Kim, Hanspeter Pfister.
DualEdit: Dual Editing for Knowledge Updating in Vision-Language Models.
- ICLR 2025** Chongjie Si*, **Zhiyi Shi***, Shifan Zhang, Xiaokang Yang, Hanspeter Pfister, Wei Shen.
Unleashing the Power of Task-Specific Directions in Parameter Efficient Fine-tuning.
- ICCV 2025** Chongjie Si, **Zhiyi Shi**, Xuehui Wang, Yichen Xiao, Xiaokang Yang, Wei Shen.
Generalized Tensor-based Parameter-Efficient Fine-Tuning via Lie Group Transformations.
- MICCAI 2024** **Zhiyi Shi**, Junsik Kim, Wanhua Li, Yicong Li, Hanspeter Pfister.
MoRA: LoRA Guided Multi-Modal Disease Diagnosis with Missing Modality.
- MICCAI 2024** Junsik Kim, **Zhiyi Shi**, Davin Jeong, Johannes Knittel, Wanhua Li, Hanspeter Pfister.
Multimodal Learning for Embryo Viability Prediction in Clinical IVF.
- Preprint 2026** Pengcheng Jiang*, Jiacheng Lin*, **Zhiyi Shi***, Heng Ji, Yejin Choi, Dawn Song, Jimeng Sun, Jiawei Han. *Adaptation of Agentic AI.*
- Preprint 2025** **Zhiyi Shi**, Junsik Kim, Helen Y. Yang, Yonghyun Song, Hyun-Jic Oh, Hanspeter Pfister.
Spatial-Temporal Pre-Training for Embryo Viability Prediction Using Time-Lapse Videos.
- Preprint 2025** **Zhiyi Shi**, Junsik Kim, Davin Jeong, Hanspeter Pfister.
Large Language Models are Zero-Shot Feature Extractors for Tabular and Text Data.

Work Experience

Parameter Efficient Fine-Tuning for LLMs

Cambridge, MA, US

Researcher at **Harvard University** (Advisor: Hanspeter Pfister)

05/2024 – 06/2025

- Provided a clear definition of task-specific directions for parameter efficient fine-tuning; designed a novel approach, LoRA-Dash, which maximizes the impact of task-specific directions during the fine-

tuning process, enhancing model performance on downstream tasks.

- Proposed Modality-aware Low-Rank Adaptation (MoRA) to disease diagnosis, which minimizes required computational resources and significantly improves performance with missing modality.
- Developed Liera, a Lie group-based PEFT method tailored for convolutional kernels, preserving their structural properties and outperforming existing approaches in vision and NLP tasks.

Project

PyHealth Library Development

06/2025 – Present

- Contributed to PyHealth of Sun Lab, an open-source deep learning library for healthcare predictive modeling (1.2k+ GitHub stars), supporting tasks such as drug recommendation, mortality prediction, and patient readmission.
- Developed the dataset loaders and preprocessing pipelines; implemented training modules and utility functions for model evaluation and improved documentation for reproducible clinical AI experiments.

Subspace Tuning Framework

08/2024 – 07/2025

- Main Contributor to a parameter efficient fine-tuning research codebase (100+ GitHub stars), integrating state-of-the-art low-rank adaptation methods and streamlining their usage with modular APIs.
- Developed modules for subspace-based extensions of LoRA and supported tasks such as language understanding, commonsense reasoning, and math reasoning, enabling standardized benchmarking across recent algorithms.

Deep Learning Library Development

09/2022 – 09/2023

- Designed and built from scratch a complete deep learning library, capable of efficient GPU-based operations, automatic differentiation of all implemented functions, and the necessary modules to support parameterized layers, loss functions, data loaders, and optimizers.
- Built several state-of-the-art modeling methods using my own deep learning library, including convolutional networks for image classification and segmentation, recurrent networks and self-attention models for sequential tasks such as language modeling, and generative models for image generation.

Academic Service

Teaching Assistant for *Deep Learning System* guided by Prof. Tianqi Chen at CMU

Reviewer for *Conference on Neural Information Processing Systems (NeurIPS)*

Reviewer for *IEEE Transactions on Artificial Intelligence*

Reviewer for *IEEE Transactions on Medical Imaging*

Reviewer for *Artificial intelligence in medicine*

Reviewer for *Computers in Biology and Medicine*

Skills

Programming & Frameworks: Python, PyTorch, TensorFlow, Hugging Face Transformers, verl.

LLMs & VLMs: Qwen2.5, LLaMA3.1, Mistral, LLaVA, GPT-4V, BLIP-2, MiniGPT4.

Fine-tuning & Adaptation: LoRA, QLoRA, DoRA, Adapter-tuning, Prefix/Prompt-tuning.

Multimodal Tasks: VQA, commonsense reasoning, cross-modal reasoning, image captioning.

Post-Training Methods: Instruct-Tuning, PPO, DPO, GRPO.

Open Source & Tools: Git, GitHub, Weights & Biases, CUDA, Linux, Slurm, Docker.

Efficiency & Scaling: Mixed precision (FP16/BF16), Quantization (INT8/4-bit), Distributed training.